# Online Learning of Kronecker Preconditioners

## Riemannian updates with two online objectives

# Matrix optimization and preconditioning

$$\min_{X \in \mathbb{R}^{m \times n}} f(X) \quad \Longleftrightarrow \quad \min_{x \in \mathbb{R}^{mn}} f(x).$$

Write $x = \mathrm{vec}(X)$ and $X = \mathrm{mat}(x)$ using column-wise vectorization. We use the same symbol $f$ for the vectorized objective.

$$d = mn, \qquad g_k = \nabla f(x_k), \qquad H(x) = \nabla^2 f(x).$$

## Kronecker structure reduces matrix costs

$$P_k = B_k \otimes A_k, \qquad A_k \in \mathbb{S}^m_{++}, \quad B_k \in \mathbb{S}^n_{++},$$

$$x_k - P_k g_k = \mathrm{vec}\big(\mathrm{mat}(x_k) - A_k \,\mathrm{mat}(g_k) B_k\big).$$

# Preconditioner selection as online learning

At round $k$, learn the factors using an online loss $\ell_k$ on

$$\mathcal{M} = \mathbb{S}^m_{++} \times \mathbb{S}^n_{++}, \qquad z = (A, B), \qquad P(z) = B \otimes A,$$

$$\mathcal{D}_{\mathrm{K}} = \{B \otimes A : A \succ 0,\ B \succ 0\}.$$

The product affine-invariant metric is

$$\langle (U, V), (U', V') \rangle_z = \operatorname{tr}(A^{-1} U A^{-1} U') + \operatorname{tr}(B^{-1} V B^{-1} V').$$

## Gradient and exponential map

For symmetric Euclidean gradients $D_A = \nabla_A \ell_k$, $D_B = \nabla_B \ell_k$,

$$\operatorname{grad} \ell_k(z) = (A D_A A,\ B D_B B).$$

$$\operatorname{Exp}_{(A,B)}(U, V) = \big(\operatorname{Exp}_A(U), \operatorname{Exp}_B(V)\big),$$

$$\operatorname{Exp}_Q(W) = Q^{1/2} \exp(Q^{-1/2} W Q^{-1/2}) Q^{1/2}.$$

# Abstract online learning framework

For fixed $0 < a_A \le b_A < \infty$ and $0 < a_B \le b_B < \infty$, let

$$\mathcal{C} = \{(A, B) \in \mathcal{M} : a_A I \preceq A \preceq b_A I,\ a_B I \preceq B \preceq b_B I\}.$$

Set $a = a_A a_B$, $b = b_A b_B$, so $aI \preceq P(z) \preceq bI$ on $\mathcal{C}$. Let $\Pi_{\mathcal{C}}$ denote the metric projection.

**Input:** $x_0$, $z_0 \in \mathcal{C}$, losses $\ell_k$, and steps $\eta_k$.

## One primal step, one learning step

**for** $k = 0, 1, \ldots$ **do**

$g_k \leftarrow \nabla f(x_k)$; **if** $g_k = 0$, **stop**.

$x_{k+1} \leftarrow x_k - P(z_k) g_k$.

Form $\ell_k(z)$ with the round-$k$ data held fixed.

$v_k \leftarrow \operatorname{grad} \ell_k(z_k)$.

$\widetilde{z}_{k+1} \leftarrow \operatorname{Exp}_{z_k}(-\eta_k v_k)$.

$z_{k+1} \leftarrow \Pi_{\mathcal{C}}(\widetilde{z}_{k+1})$.

**end for**

The learned $P_{k+1}$ is used in the next primal step.

# Objective II: LogDet curvature matching

Let $H_k = \nabla^2 f(x_k) \succ 0$ and $P = B \otimes A$. Define

$$\ell_k^{\mathrm{LD}}(A, B) = \frac{1}{d}\big[\mathrm{tr}(H_k P) - \log\det P\big].$$

This loss is geodesically convex on $\mathcal{M}$. Its unrestricted matrix minimizer is $H_k^{-1}$.

## Exact Riemannian gradient

Set $E = P^{1/2} H_k P^{1/2} - I_d$, where $P^{1/2} = B^{1/2} \otimes A^{1/2}$. Then

$$\Omega_A = \frac{1}{d}\,\mathrm{Tr}_n E, \qquad \Omega_B = \frac{1}{d}\,\mathrm{Tr}_m E,$$

$$\mathrm{grad}\,\ell_k^{\mathrm{LD}}(z) = \big(A^{1/2}\Omega_A A^{1/2},\ B^{1/2}\Omega_B B^{1/2}\big).$$

For $E = [E_{ij}]_{i,j=1}^n$ in $m \times m$ blocks, $\mathrm{Tr}_n E = \sum_i E_{ii}$ and $[\mathrm{Tr}_m E]_{ij} = \mathrm{tr}(E_{ij})$.

Exact contractions generally require full Hessian information or many Hessian–vector products (HVPs).

Geometric background: Wiesel (2012), Geodesic Convexity and Covariance Estimation.

# LogDet learning with a single random probe

Draw $Z \in \{-1, +1\}^{m \times n}$ with fresh independent, equiprobable entries.
Let $\xi = \mathrm{vec}(Z)$, so $\mathbb{E}[\xi\xi^\top] = I_d$.

## One HVP estimates both gradient components

$$u = \mathrm{vec}(A^{1/2} Z B^{1/2}), \qquad v = H_k u,$$

$$R = A^{1/2}\,\mathrm{mat}(v) B^{1/2} - Z, \qquad \mathrm{vec}(R) = E\xi,$$

$$\widehat{\Omega}_A = \frac{\mathrm{sym}(RZ^\top)}{d}, \qquad \widehat{\Omega}_B = \frac{\mathrm{sym}(R^\top Z)}{d}, \qquad \mathrm{sym}(C) = \tfrac{1}{2}(C + C^\top).$$

$$\widehat{v} = \left(A^{1/2}\widehat{\Omega}_A A^{1/2},\ B^{1/2}\widehat{\Omega}_B B^{1/2}\right).$$

For the history $\mathcal{F}_k$ before the fresh probe,

$$\mathbb{E}[\widehat{v}_k \mid \mathcal{F}_k] = \mathrm{grad}\,\ell_k^{\mathrm{LD}}(z_k).$$

No full Hessian is formed. If $P = H_k^{-1}$, then $R = 0$ for every probe.

# Algorithm II: stochastic LogDet learning

**Input:** $x_0$, $z_0 \in \mathcal{C}$, $L$, and rates $\eta_k > 0$. Set $\alpha_s = (Lb)^{-1}$.

## Safeguarded primal step and one-probe learning

**for** $k = 0, 1, \ldots$ **do**

$g_k \leftarrow \nabla f(x_k)$; **if** $g_k = 0$, **stop**.

$d_k \leftarrow \mathrm{vec}(A_k \,\mathrm{mat}(g_k) B_k)$.

$\widehat{x}_k \leftarrow x_k - d_k, \quad x_k^s \leftarrow x_k - \alpha_s d_k$.

$x_{k+1} \in \underset{y \in \{\widehat{x}_k, x_k^s\}}{\arg\min}\ f(y)$.

Draw a fresh $Z_k$ and compute $\widehat{\Omega}_{A,k}, \widehat{\Omega}_{B,k}$
using the preceding one-HVP formulas at $x_k, z_k$.

$\widetilde{A}_{k+1} \leftarrow A_k^{1/2} \exp(-\eta_k \widehat{\Omega}_{A,k}) A_k^{1/2}$.

$\widetilde{B}_{k+1} \leftarrow B_k^{1/2} \exp(-\eta_k \widehat{\Omega}_{B,k}) B_k^{1/2}$.

$z_{k+1} \leftarrow \Pi_{\mathcal{C}}(\widetilde{A}_{k+1}, \widetilde{B}_{k+1})$.

**end for**

The HVP uses $\nabla^2 f(x_k)$. Both primal candidates use $P_k$.

# Assumptions and notation for convergence

## Global strong convexity and smoothness

Assume $f \in C^2(\mathbb{R}^d)$ and

$$\mu I_d \preceq H(x) = \nabla^2 f(x) \preceq L I_d \quad \text{for every } x, \qquad 0 < \mu \leq L.$$

Let $x_\star$ be the unique minimizer and define

$$H_\star = \nabla^2 f(x_\star), \qquad \Delta_k = f(x_k) - f(x_\star).$$

Initialize $z_0 \in \mathcal{C}$, so $aI \preceq P_k \preceq bI$.

## Additional regularity for learning limits and superlinear rates

Assume the Hessian is Lipschitz near $x_\star$: for some $M_H, r > 0$,

$$\|H(x) - H(y)\|_{\mathrm{op}} \leq M_H \|x - y\| \quad \text{for } x, y \in B(x_\star, r).$$

# Convergence of stochastic LogDet learning

## Theorem: global linear convergence

Under the global assumptions, for every realization of the probes,

$$\Delta_{k+1} \leq \left(1 - \frac{\mu a}{L b}\right) \Delta_k.$$

## Theorem: almost-sure preconditioner convergence

Assume also local Hessian Lipschitzness. Let $\mathcal{C}$ contain a factor representation of the global LogDet minimizer over $\mathcal{D}_{\mathrm{K}}$. For deterministic rates satisfying

$$\eta_k > 0, \qquad \sum_k \eta_k = \infty, \qquad \sum_k \eta_k^2 < \infty,$$

the fresh independent probes give, along any infinite run,

$$P_k \longrightarrow P_{\mathrm{K}} = \operatorname*{arg\,min}_{P \in \mathcal{D}_{\mathrm{K}}} \{\operatorname{tr}(H_\star P) - \log\det P\} \quad \text{a.s.}$$

The optimal matrix $P_{\mathrm{K}}$ is unique.

# Superlinear convergence of stochastic LogDet learning

## Corollary under exact Kronecker structure

Under the preceding learning theorem, suppose

$$H_\star^{-1} = B_\star \otimes A_\star, \qquad (A_\star, B_\star) \in \mathcal{C}.$$

Then $P_{\mathrm{K}} = H_\star^{-1}$ and, along any infinite run,

$$P_k \longrightarrow H_\star^{-1} \quad \text{almost surely,}$$

$$\frac{\|x_{k+1} - x_\star\|}{\|x_k - x_\star\|} \longrightarrow 0, \qquad \frac{\Delta_{k+1}}{\Delta_k} \longrightarrow 0 \quad \text{almost surely.}$$

The algorithm learns the entire structured inverse Hessian using one random probe and one HVP per round.

Finite termination solves the primal problem exactly. It need not mean that the saved preconditioner already equals $H_\star^{-1}$.

# Objective I: the function value after a trial step

Freeze $x_k$ and $g_k$. For $g_k \neq 0$, define

$$h_k(A, B) = \frac{f(x_k - (B \otimes A)g_k) - f(x_k)}{\|g_k\|^2}.$$

At $z_k$, set

$$\widehat{x}_k = x_k - P_k g_k, \qquad G_k = \mathrm{mat}(g_k), \qquad T_k = \mathrm{mat}\big(\nabla f(\widehat{x}_k)\big).$$

With $c_k = \|g_k\|^2$,

$$D_{A,k} = -\frac{\mathrm{sym}(T_k B_k G_k^\top)}{c_k},$$

$$D_{B,k} = -\frac{\mathrm{sym}(G_k^\top A_k T_k)}{c_k},$$

$$\mathrm{grad}\, h_k(z_k) = (A_k D_{A,k} A_k,\ B_k D_{B,k} B_k).$$

- One additional gradient evaluation at the trial point.
- No Hessian or random gradient estimate is required.

Related feedback: Gao et al. (2025), Online Scaling, Part I.

# Algorithm I: deterministic function-value learning

**Input:** $x_0$, $z_0 \in \mathcal{C}$, a smoothness constant $L$, and a fixed learning rate $\eta > 0$.

## Safeguarded primal step and exact factor learning

**for** $k = 0, 1, \ldots$ **do**

$g_k \leftarrow \nabla f(x_k)$; **if** $g_k = 0$, **stop**.

$\widehat{x}_k \leftarrow x_k - P_k g_k$, where $P_k = B_k \otimes A_k$.

$$x_{k+1} \in \operatorname*{arg\,min}_{y \in \{\widehat{x}_k,\, x_k - g_k/L\}} f(y).$$

$G_k \leftarrow \mathrm{mat}(g_k), \quad T_k \leftarrow \mathrm{mat}\big(\nabla f(\widehat{x}_k)\big)$.

Compute $D_{A,k}, D_{B,k}$ from the feedback formulas.

$v_k \leftarrow (A_k D_{A,k} A_k,\ B_k D_{B,k} B_k)$.

$\widetilde{z}_{k+1} \leftarrow \mathrm{Exp}_{z_k}(-\eta v_k)$.

$z_{k+1} \leftarrow \Pi_{\mathcal{C}}(\widetilde{z}_{k+1})$.

**end for**

The feedback always evaluates the preconditioned trial, even when the safeguard accepts the ordinary gradient step.

# Convergence of deterministic function-value learning

## Theorem: global linear convergence

Under the global assumptions, for any fixed $\eta > 0$,

$$\Delta_{k+1} \leq \left(1 - \frac{\mu}{L}\right) \Delta_k.$$

Thus the function gaps are Q-linear and the iterates are R-linear.

## Theorem: local superlinear rate

Assume also local Hessian Lipschitzness and

$$H_\star^{-1} = B_\star \otimes A_\star, \qquad (A_\star, B_\star) \in \mathcal{C}.$$

For a sufficiently small fixed $\eta > 0$, either the algorithm terminates finitely or

$$\frac{\|x_{k+1} - x_\star\|}{\|x_k - x_\star\|} \longrightarrow 0, \qquad \frac{\Delta_{k+1}}{\Delta_k} \longrightarrow 0.$$

No geodesic convexity of $h_k$, or convergence of the entire $P_k$ to $H_\star^{-1}$, is required.